Pratik Koppikar

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EDUCATION

University of Cambridge August 2023 – Present
Doctor of Philosophy in Biochemistry | King's College Cambridge
Supervisors: Michael Ward, Florian Merkle, Andrew Bassett
Funding: National Institutes of Health Oxford-Cambridge Scholars Programme (NIH OxCam), Cambridge Trust

The University of Texas at Dallas August 2019 – May 2023
Bachelor of Science in Biology | Minors: Political Science, Secondary STEM Education **GPA: 4.0**
Honors: Eugene McDermott Scholars Program, Summa Cum Laude, National Merit Scholars Program, Collegium V Honors College, Phi Kappa Phi, Major Honors

EXPERIENCE

Flagship Pioneering May 2026 – August 2026
Fellow *Boston, MA*

- Created novel scientific venture hypotheses by analyzing emergent paradigms across computational microscopy, physics constraints on drug discovery/development, and multimodal world models.
- Evaluated multidisciplinary research to assess scientific plausibility and competitive landscapes for stealth-stage ventures.
- Pitched validated venture concepts to Flagship partners and leadership.

National Institutes of Health & University of Cambridge August 2023 – Present
Doctoral Student / NIH OxCam Scholar *Bethesda, MD / Cambridge, UK*

- Optimization of base and prime gene editing tools for use in human induced pluripotent stem cell (hiPSC) model systems to effect pathogenic Alzheimer's Disease and Related Dementias (ADRD) variants.
- Segmentation and base classification of large-scale optical pooled screening (OPS) datasets to extract cell-level phenotypes and barcode identities from multiplexed imaging and sequencing data.
- Regression and neural network-based modeling to predict gene identity from microscopy images of endogenously tagged fluorescent proteins, using data from proteome-wide pooled next-gen editing.
- Molecular dynamics simulation of tagged proteins for computational-experimental looped validation of intrinsically favorable internal tag regions.

Revvity (formerly PerkinElmer Genomics) May 2021 – May 2023
Research Intern *Pittsburgh, PA*

- Analytical chemistry to produce 3 assays/protocols translated to clinical production, including for Facioscapulohumeral Muscular Dystrophy through Optical Genome Mapping.
- Developed liquid chromatography-tandem mass spectrometry assay for quantification of Cerebrotendinous xanthomatosis bile acid analytes CDCA and THCA.
- RNA extraction and library preparation protocols from fresh blood and frozen tissue.

Yale School of Medicine May 2022 – September 2022
Research Intern, Monkol Lek Group *New Haven, CT*

- Single-cell atlas for muscle cells with representation of histological features.
- Dimensionality reduction, high-throughput image analysis, and web portal development.

Center for Life Sciences at National University of Singapore January 2022 – May 2022
Research Intern, Jai Polepalli Group *Singapore*

- Validated use of Cre-lox and Flp-frt genetic manipulation tools using cloning, transduction, and cell culture.

- Tested specificity and suitability of target antibody to the 5HT3AR interactor protein found in axons and nerve terminals.
- Designed co-immunoprecipitation methodology for synaptosome preparation and quantification of target interactor protein.

Potomac Institute for Policy Studies August 2021 – December 2021
Science and Technology Policy Intern *Washington, D.C.*

- Developed a capstone analytical review on the state of government funding for neural implantable technology.
- Drafted weekly analytical briefs on emergent technologies including smart cities, gene editing, and data privacy.

Neuronal Networks and Interfaces Lab at UT Dallas October 2019 – May 2021
Research Assistant, Joseph Pancrazio Group *Dallas, TX*

- Investigated axonal nociceptive function in sensory neurons through use of microelectrode arrays integrated with microfluidics.
- Assessed the role of Ca-dependent K channels in dorsal root ganglion (DRG) stimulation through analysis of electrical stimulation data from microelectrode arrays.
- Quantified colocalization of DRGs with nociceptive marker TRPV1 from representative immunocytochemistry images, identifying a 2.7x increased axonal expression after IL-6 sensitization.

Human Genetics Lab at Emory University June 2018 – August 2018
Research Assistant, Madhuri Hegde Group *Atlanta, GA*

- Tested non-invasive alternative diagnosis pathway for limb-girdle muscular dystrophies (LGMDs) utilizing whole blood cells.
- Performed and analyzed western blot results to verify CAPN3 as a major contributing gene to the LGMD phenotype.
- Validated enzyme assays for GAA enzyme related to Pompe disease from muscle biopsy and fresh blood protein extract.

PUBLICATIONS

Guruju, N. M., Jump, V., Lemmers, R., Van Der Maarel, S., Liu, R., Nallamilli, B. R., Shenoy, S., Chaubey, A., **Koppikar, P.**, Rose, R., Khadilkar, S., & Hegde, M. (2023). Molecular Diagnosis of Facioscapulohumeral Muscular Dystrophy in Patients Clinically Suspected of FSHD Using Optical Genome Mapping. *Neurology: Genetics*, 9(6), e200107. doi:10.1212/NXG.0000000000200107

Koppikar, P., Shenoy, S., Guruju, N., & Hegde, M. (2023). Testing for Facioscapulohumeral Muscular Dystrophy with Optical Genome Mapping. *Current Protocols*, 3(1), e629. doi:10.1002/cpz1.629

Atmaramani, R., Veeramachaneni, S., Mogas, L. V., **Koppikar, P.**, Black, B. J., Hammack, A., Pancrazio, J. J., & Granja-Vazquez, R. (2021). Investigating the function of adult DRG neuron axons using an in vitro microfluidic culture system. *Micromachines*, 12(11), 1317. doi:10.3390/mi12111317

PRESENTATIONS & TALKS

Invited & Selected Oral Presentations

- **Koppikar, P.** "Visualizing the Unseen: High-Throughput Optical Pooled Screening and Machine Learning in hiPSCs." *King's College Graduate High Table Seminar*, King's College, Cambridge, UK. February 2026
- Shin, J., **Koppikar, P.**, & Kwok, K. "Maintaining Security in Space Through Enhanced Verification & Attribution Capabilities: An Integrated Technical and Policy Approach." *77th International Astronautical Congress (IAC)*, Antalya, Türkiye. October 2026

Selected Poster Presentations

- **Koppikar, P.**, Bakavayev, S., Yuan, H., & Ward, M. E. "Characterization of AD-Associated Variant Effectuation Efficiency by Base and Prime Editors in iPSC Models." *Packard Center ALS Research Symposium.* 2025
- **Koppikar, P.**, et al. "Scalable Endogenous Tagging by Pooled Prime Editing for Characterizing Protein Localization." *Genomics at Scale Conference.* 2025
- **Koppikar, P.**, et al. "Scalable Endogenous Tagging by Pooled Prime Editing for Characterizing Protein Localization." *University of Cambridge Department of Pharmacology Away Day*, Cambridge, UK. 2025

AWARDS & FELLOWSHIPS

Lili Sarnyai Graduate Research Prize, King's College Cambridge 2026
Selected as top presentation at the Graduate Research Showcase at King's College Cambridge.

Cambridge Trust Scholarship 2023
Full doctoral tuition and stipend support for study at the University of Cambridge.

NIH Oxford-Cambridge Scholars Fellowship 2023
Dual-mentorship international doctoral fellowship between NIH and Cambridge.

Barry Goldwater Scholarship 2022
Premier undergraduate award in the United States for mathematics, natural sciences, and engineering.

Bill Archer Fellowship 2021
Competitively selected public policy fellowship in Washington, D.C., through the University of Texas System.

Eugene McDermott Scholars Program 2019 – 2023
Full academic merit scholarship, research funding, and travel endowment at UT Dallas (<20 awarded annually).

Eagle Scout, Boy Scouts of America 2019

TEACHING & MENTORSHIP

Research Supervision

- **Imran Bahiru** (National Institutes of Health) June 2026 – August 2026
  Project: *Designing a Binder for TMEM106B.*
  Outcome: Mentored in bash scripting, BindCraft de novo protein design workflows, and structural analysis.
- **Elizabeth Chew** (University of Cambridge) January 2026 – May 2026
  Project: *Establishing a Split-mNeonGreen2 Complementation Assay to Map Ciliary GPCR Localisation in mIMCD3 Cells.*
  Outcome: Mentored in ciliogenesis protocols, widefield fluorescence imaging, and semi-automated Fiji pipelines.

Teaching & Academic Service

- **Student Teacher**, Garland High School January 2023 – May 2023
  Co-taught and assisted in classroom instruction, grading, and laboratory modules for 9th-grade Pre-AP Biology and ESL Biology under mentor teacher Maria Azhar (150+ students).
- **Peer Tutor / Mentor**, Eugene McDermott Scholars Program 2020 – 2023
  Mentored incoming undergraduate STEM scholars through academic planning and fellowship applications.

COMPUTATIONAL PROJECTS

Mutation-Aware Latent Activations 2025 – 2026

- Built a sparse autoencoder (SAE) of ESM2-650M layer 24, generating Δ-heat-maps comparing wild-type and mutant latent activation disparities across per-residue sequences.
- Uncovered distinct activation patterns at functional protein motifs and termini localized to mutation sites.

- Identified interpretable mutation-aware latent features indicative of structural instability and variant pathogenicity.

SKILLS & INTERESTS

Computational: Python, PyTorch Lightning, Docker, Snakemake, Bash, Git, HPC (Slurm), Java, AMBER, Jupyter
Wet Lab: hiPSC culture, neuronal culture, mRNA synthesis, functional genomics (Perturb-seq, OPS), cloning, transgenic delivery (lentivirus, LNP), optical genome mapping, LC-MS/MS
Interests: Jazz (guitar, vibraphone), 20th-century Russian and Japanese literature, Dallas Mavericks basketball